Justin Amen N'GBLOGNI

Final-Year Engineering Internship — Mechatronics & Robotics (M.Eng. Year 5) | *March 6 – September 6, 2027*

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PROFESSIONAL SUMMARY

Final-year engineering student (M.Eng., ENSIL-ENSCI / ISAE-Supméca) specializing in **perception and control for autonomous systems**. At Politecnico di Torino I built a complete robotic navigation stack — from embedded semantic perception to GPU-accelerated predictive control — and took it from mathematical modeling to validation on physical hardware. Seeking a 6-month final-year internship in robotics, perception, or control engineering at a major industrial organization. Open to international relocation.

EDUCATION

M.Eng. Year 5, Exchange Program — Mechatronics & Complex Systems — 2026 - 2027
ISAE-Supméca, Graduate School of Mechanical Engineering — *Paris, France*

M.Eng. — Mechatronics, Robotics & Automation (5-year integrated program) — 2024 - 2027
ENSIL-ENSCI, Graduate School of Engineering (French Grande École) — *Limoges, France*

Integrated Preparatory Program — Electrical Engineering — 2021 - 2024
ENSIL-ENSCI (France) / École Polytechnique d'Abomey-Calavi (Benin) — *France / Benin*

Scientific Baccalaureate — Highest Honors (Mention Très Bien) — 2021
St Augustin Secondary School — *Benin*

PROFESSIONAL EXPERIENCE

R&D Intern — Perception & Risk-Aware Autonomous Navigation — May 2026 - July 2026
Politecnico di Torino, DIMEAS Department (advisors: Prof. F. F. Lizzio, Prof. E. Capello) — *Turin, Italy*

- Built an end-to-end navigation stack for a TurtleBot 4: a two-layer risk map (RGB-D semantic segmentation + geometric depth) driving a GPU-accelerated MPPI controller evaluating 5,120 trajectories per cycle at 14 Hz.
- Results: cut risk exposure by 44% under forced traversal and by 100% in avoidable zones, at only 2-9% overhead in distance and travel time. Benchmarked and selected YOLO26 for embedded deployment (approx. 46 FPS on-device vs. 7-10 FPS CPU-only).
- Derived a cost-arbitration model predicting the exact threshold at which the robot switches from traversing to circumnavigating a hazard; validated against physical measurements, then corrected and re-validated. Stack: ROS 2 Jazzy, Gazebo Harmonic, C++/CUDA, Python, PyTorch.

Assembly Line Operator Intern — June 2025 - July 2025
Stellantis Poissy — Stellantis Group (formerly PSA) — *Poissy, France*

- Final assembly of Opel Mokka and DS 3 Crossback vehicles: hands-on exposure to takt time, quality standards, and safety protocols on a high-volume production line.

Robotics & AI Instructor — 2022 - 2025
Fondation Vallet (Fondation de France) — EEIA Summer School — *Benin*

- Designed and led hands-on workshops in AI and embedded robotics for audiences ranging from middle-school students to working professionals.

TECHNICAL PROJECTS

FlyRenov — Redesign of a Servo-Controlled Hose Winder for a Roof-Cleaning Drone — 2025 - 2026
Industrial project, ENSIL-ENSCI × FlyRenov — *France*

- Full machine redesign with no loss of functionality: 3x lighter (40 to 14 kg / 88 to 31 lb, now single-person portable), 4 motors reduced to 2, and an entire CAN network replaced by a single ESP32 board.
- Closed-loop tension and length control of a 60 m tethered hose (VESC/FOC, UART); DFMA-oriented design, CNC machining, digital mock-up in 3DEXPERIENCE.

Autonomous Robotic Arm — Tic-Tac-Toe via Reinforcement Learning — 2025
EEIA Summer School — *Benin*

- Deep reinforcement learning agent (DQN/PPO) reaching 98% accuracy on complete games, with YOLO-based visual board detection and real-time inference embedded on a Raspberry Pi 5.

Autonomous Vehicle — From Kit to Ground-Up Design — 2021 - 2024
EEIA Summer School — *Benin*

- From-scratch redesign of the vehicle (CAD in Fusion 360, 3D printing, electronics, software) with self-driving via imitation learning, from distance-sensor guidance to camera vision (TensorFlow, OpenCV, TinyML).

TECHNICAL SKILLS

AI & Computer Vision:
PyTorch, TensorFlow, OpenCV, semantic segmentation (SegFormer, MobileNet), YOLO, reinforcement learning, TinyML.

Robotics & Perception:
ROS 2 (Jazzy), Gazebo Harmonic, NVIDIA Isaac Sim, MPPI & model predictive control, SLAM, path planning, Kalman filtering, sensors (IMU, GPS, LiDAR, OAK-D RGB-D camera).

Programming & Embedded:
Python, C, C++, CUDA, Arduino, ESP32, STM32, RTOS, Raspberry Pi, embedded Linux, AI-assisted development (LLMs). Protocols: CAN, SPI, I2C, UART.

Design & R&D Tools:
MATLAB/Simulink, Simcenter Amesim, SolidWorks, Fusion 360, 3DEXPERIENCE, EasyEDA, Altium Designer, KiCad, Git/GitLab.

AWARDS, LANGUAGES & INTERESTS

Awards & Certifications: Fondation Vallet Excellence Scholarship (2023-2027) • 2nd National Prize, EEIA (2021) • MathWorks (MATLAB, Simulink) • Arduino
Languages: French (Native) • English (Intermediate, CEFR B1) • Italian (Intermediate, CEFR B1) • German (Beginner, A1)
Interests: Automotive, Robotics & AI, Travel, Drawing, Music, Cooking.